



Ankit Kumar

Computational Drug Discovery Scientist | Cheminformatics & ML

🏠 Troisdorf, Germany ✉ chem.ak91@gmail.com ☎ +49 176 87248106

🌐 LinkedIn 🐙 GitHub 🌐 Portfolio

Professional Profile

Computational drug discovery scientist with over seven years of experimental research experience with hands-on expertise in machine learning, cheminformatics (RDKit), molecular modeling (Maestro, LiveDesign), and molecular dynamics (GROMACS). Develops predictive models and analyze molecular data to accelerate drug discovery. Applies AI-driven approaches in molecular design, digital chemistry, and lab automation.

Professional Experience

AI Trainer

Mar 2025–Present

Outlier.ai | Troisdorf, Germany (Remote)

- Create and curate high-quality datasets and evaluation frameworks for STEM-focused AI models and drug discovery agents
- Review, edit, and improve work submitted by peers to align with client requirements and quality standards
- Provide feedback on chemical data accuracy and scientific validity for AI training datasets

Career Transition: AI & Machine Learning in Chemistry

Apr 2024–Feb 2025

Self-Directed Learning | Troisdorf, Germany

- Self-study in machine learning fundamentals, deep learning, and AI applications in chemistry
- Developed expertise in cheminformatics using RDKit for molecular property prediction and QSAR modeling
- NVIDIA-Certified Associate: Generative AI LLMs; achieved B1 German language certification

Data Science Trainee

Nov 2023–Mar 2024

neuefische GmbH — School for Digital Talent | Hamburg, Germany (Remote)

- Completed rigorous bootcamp covering machine learning algorithms, statistical modeling, and data science workflows
- Developed end-to-end ML projects including data collection, preprocessing, model training, and evaluation
- Gained proficiency in model performance metrics, cross-validation, and hyperparameter optimization

Full-Time Parenting

Jul 2022–Oct 2023

Troisdorf, Germany

Postdoctoral Research Associate

Dec 2020–Jun 2022

Technical University Dresden | Dresden, Germany

- Synthesized and analyzed carbon-based materials such as graphene nanoribbons and spin chains
- Developed scalable methodologies for hetero-structure graphene nanoribbons for optoelectronic applications
- Maintained lab records, managed equipment, and prepared comprehensive project reports

Senior Research Associate

Jan 2020–Oct 2020

Neogen Chemicals Limited | Vadodara, India

- Developed scalable methods for mass production of pharmaceutical intermediates
- Achieved 10–15% yield improvements, reducing production costs and enhancing profitability
- Interpreted NMR, IR, and UV data, maintained laboratory records, and prepared reports

Doctor of Philosophy (Ph.D.) in Chemistry

Jul 2014–Dec 2019

Indian Institute of Technology Bombay | Mumbai, India

- Developed polyaromatic hydrocarbon based porphyrinoid macrocycles and metal complexes
- Strong experience with NMR, HR-MS, UV–Vis, fluorescence, electrochemistry, and X-ray crystallography

- Published 13 peer-reviewed articles in high-impact journals

Education

Doctor of Philosophy (Ph.D.) in Chemistry

Indian Institute of Technology Bombay, Mumbai, India | 2014–2019

Master of Science (M.Sc.) in Chemistry

CCS University Meerut, India | 2010–2012

Key Data Science Projects

AqSolPrediction: Aqueous Solubility Prediction — [GitHub](#)

- Built ML regression pipeline on AqSolDB (~10k compounds) using RDKit molecular descriptors and Morgan Fingerprints, achieving test $R^2 = 0.88$
- Deployed interactive web app on Hugging Face Spaces with real-time SMILES input and prediction
- Implemented feature engineering with molecular fingerprints and physicochemical property calculations

Tech: Python, RDKit, Hugging Face, Docker, FastAPI

AgronomiCare: Plant Disease Prediction — [GitHub](#)

- Developed ML web application for agricultural disease prediction using classification algorithms
- Implemented complete data pipeline: data collection, cleaning, feature extraction, model training, deployment

Tech: Python, Machine Learning, Web App

Kickstarter Success Prediction — [GitHub](#)

- Built multi-model classification system comparing logistic regression, KNN, XGBoost, and Random Forest
- Performed extensive feature engineering and data preprocessing on real-world business dataset

Tech: Python, XGBoost, Random Forest, Data Analysis

Certifications & Publications

- Schrödinger – Designing Quality Ligand Libraries
- MD Simulations & Computational Biophysics (GROMACS, VMD, ChimeraX) – BDGLifeSciences
- Schrödinger – Introduction to Molecular Modeling in Drug Discovery
- NVIDIA Certified – Generative AI & LLMs
- Data Science Bootcamp – neuefische GmbH
- Author of 13 peer-reviewed scientific publications in prestigious international journals. [[Google Scholar](#)]

Skills

AI & ML: Machine Learning, Deep Learning, Predictive Modeling, Python, Pandas, NumPy, Pytorch, Scikit-Learn

Computational Chemistry: Molecular Docking, Molecular Dynamics (GROMACS, VMD, ChimeraX), Ligand Library Design, CADD

Cheminformatics: RDKit, QSAR Modeling, Molecular Modeling, Molecular Property Prediction

R&D: Organic Synthesis, Process Chemistry, Yield Optimization, Scale-Up

Analytical: NMR, FT-IR, UV-Vis, Fluorescence, HPLC, GC-MS, Mass Spectrometry, Chromatography

Tools and Platforms: Maestro, LiveDesign, PubChem, ChEMBL, Docker, MLFlow, SQL, FastAPI, Git, VS Code

Languages

German: B1 Certified **English:** Full Professional Proficiency **Hindi:** Native

Additional Information

Work Authorization: Valid work permit for Germany; eligible to work for German companies